Ahamed Mohamed Wajibu

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EDUCATION

UNIVERSITY OF WATERLOO — Waterloo, ON
Honours Bachelor of Mathematics — Expected May 2028

SKILLS / TECHNOLOGIES

- **Languages:** C++, Rust, CUDA, Python, Go, C, C#, TypeScript, JavaScript, WGSL
- **ML & Inference:** PyTorch, TensorFlow, cuBLAS, OpenCV, YOLO, LangChain, Ollama
- **Backend & Cloud:** ASP.NET Core, FastAPI, Flask, Node.js, React, SQL Server, PostgreSQL, Docker, Nginx, Linux, Git, Azure, AWS, Microsoft Fabric

EXPERIENCE

RBC Borealis — Toronto, ON
*AI Researcher Fellow | **1 of 10** teams selected across Canada* — *October 2026 – Present*

- Building a computer vision pipeline (YOLO, ByteTrack/DeepSORT) on the Kaggle NFL 1st and Future impact-detection data to detect and track players and helmets across multi-camera footage and attribute head impacts.
- Developing PyTorch temporal models over player-tracking features (relative velocity, acceleration, collision angle, orientation) to classify impacts and estimate impact magnitude and per-athlete exposure risk.

Mitsubishi — Mississauga, ON
AI Engineer Intern — *January 2026 – May 2026*

- Owned an internal RAG platform end to end over 2,000+ aircraft technical manuals, serving **40 engineers** (LangChain, Azure AI Foundry, PostgreSQL, Microsoft Fabric); ingestion pipelines extract structured fields, per-page summaries, and source metadata.
- Answered **98 of 100** questions correctly on a golden set written by senior engineers; both misses required image understanding.
- Self-hosted Gemma 4 and quantized open-source models via Ollama behind an Nginx reverse proxy on an RTX PRO 6000 (Blackwell) server, keeping sensitive documents on-prem.
- Added tool calling (document search, page retrieval, SQL keyword search, web search) with inline citations.

Mitsubishi — Mississauga, ON
Software Developer Intern — *September 2025 – December 2025*

- Replaced manual SQL Server scripts with ASP.NET Core/C# pages for QA call-up, aircraft line status, job updates, and Syspro service workflows used by hundreds of production workers.
- Refactored Entity Framework services, LINQ queries, and SQL Server access patterns, cutting request latency by **~30%** and reducing redundant database calls across production modules.

PROJECTS

Forge | *Rust, C++, CUDA, Go | GitHub*

- Built an LLM inference runtime **from scratch** for Qwen2.5-0.5B and SmolLM2-135M: transformer decode, KV cache, RoPE, RMSNorm, and attention kernels in C++/CUDA, a Rust runtime, and an OpenAI-compatible streaming server in Go.
- Implemented **continuous batching**, **a paged KV cache**, and **CUDA Graphs** for GPU serving; on an NVIDIA L4, aggregate decode throughput scales from 22.0 to 41.6 tok/s across batch sizes 1 to 8.
- Raised CPU decode from **0.44 to 23.0 tok/s (~52×)** through multithreaded kernels and weight-only INT8 quantization (group size 64), which also cut memory use from 2,404 to 1,020 MiB.

Block-GTQ WebGPU | *TypeScript, WGSL, WebGPU, PyTorch | GitHub*

- Ported Block-GTQ's RoPE-aware K-cache quantization (Liang et al.) to WebGPU/WGSL, with mixed-width bit allocation, bit-packed storage, and in-kernel key decompression; runs SmolLM2-135M and Qwen2.5-0.5B in the browser.
- Cut persistent K-cache memory **2.24×** at 4K context (45.35 → 20.24 MiB) and validated GPU attention against a Python reference across **120 strict comparisons** (max error 2.4e-6).
- Removed full-K reconstruction from compressed attention, cutting scratch memory from 2.20 MB to 101 KB and attention time **22×** versus my first implementation (4,096×128 benchmark); the compressed path remains slower end to end, as documented in the repo.